

NEWARK-HEATH, OH, USA

WMO: 725229

Lat: 40.023N

Lon: 82.463W

Elev: 884

StdP: 14.23

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 04858

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 1.2	(c) 7.8	(d) -6.2	(e) 4.1	(f) 2.8	(g) -0.6	(h) 5.5	(i) 9.7	(j) 25.0	(k) 36.0	(l) 22.8	(m) 33.5	(n) 4.4	(o) 250	(p) 0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	89.9	73.8	87.8	72.8	85.5	71.5	76.8	86.0	75.3	83.5	73.9	81.5	8.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.7	129.6	81.5	72.5	124.5	79.9	71.6	120.7	78.6	40.7	86.1	39.2	83.9	37.9	81.2	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.6	(b) 17.9	(c) 15.8	(e) -7.9	(f) 92.9	(g) 8.6	(h) 2.7	(i) -14.1	(j) 94.8	(k) -19.1	(l) 96.4	(m) -24.0	(n) 97.9	(o) -30.3	(p) 99.8		
(4) 20.6	(5) 17.9	(6) 15.8	(7) -8.4	(8) 79.1	(9) 8.4	(10) 1.6	(11) -14.5	(12) 80.2	(13) -19.4	(14) 81.1	(15) -24.2	(16) 82.0	(17) -30.3	(18) 83.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.2	28.2	30.6	40.6	52.4	62.3	70.3	73.2	71.6	65.4	53.6	42.6	34.1
	DBStd	18.17	12.73	11.62	10.87	8.97	8.27	5.78	5.03	4.89	6.86	8.88	9.00	9.88
	HDD50	2467	681	550	332	79	8	0	0	0	1	61	256	499
	HDD65	5507	1140	965	756	385	154	20	3	7	78	369	673	957
	CDD50	3265	6	6	41	150	388	608	720	671	462	173	33	7
	CDD65	828	0	0	1	6	69	178	258	212	88	16	0	0
	CDH74	7688	0	1	15	114	701	1616	2260	1861	925	191	4	0
	CDH80	2392	0	0	2	11	176	517	777	575	291	43	0	0
Wind	WSAvg	5.8	7.7	7.2	7.0	7.3	5.7	5.0	4.0	3.6	3.8	5.2	6.1	6.8
Precipitation	PrecAvg	40.6	3.1	2.3	3.2	3.6	4.2	4.3	4.2	3.5	3.2	2.9	2.8	3.1
	PrecMax	54.4	8.4	4.6	7.3	5.8	7.7	9.6	10.1	7.2	6.8	5.6	5.1	8.1
	PrecMin	29.1	1.1	0.9	1.1	1.6	1.1	0.8	1.6	1.4	1.2	1.1	1.0	1.2
	PrecStd	5.6	1.7	0.9	1.4	1.0	1.9	2.1	2.0	1.7	1.3	1.3	1.0	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.6	65.9	75.6	81.6	88.2	91.3	92.7	91.3	90.8	85.5	72.1	63.9
		MCWB	58.0	56.9	61.9	64.1	70.5	72.7	76.8	74.2	72.4	70.8	61.1	57.4
	2%	DB	57.7	58.8	69.6	77.4	84.5	88.3	89.5	88.4	87.3	79.1	66.4	58.5
		MCWB	54.1	52.3	57.8	61.4	68.5	71.9	75.1	73.8	71.7	65.7	56.9	53.7
	5%	DB	52.4	53.8	63.6	73.4	81.6	85.8	87.3	85.8	83.1	74.5	62.7	54.1
		MCWB	48.8	48.4	54.3	59.3	67.9	70.6	73.6	72.5	69.7	62.7	54.9	50.0
	10%	DB	46.3	47.9	59.0	69.5	78.2	82.8	84.5	83.2	79.6	69.9	58.6	49.6
		MCWB	42.1	42.9	52.0	57.0	66.1	69.6	72.2	71.5	67.4	61.0	51.8	45.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.6	59.1	63.2	67.5	73.4	76.5	79.1	78.3	75.7	72.0	63.6	59.1
		MCDB	62.6	65.2	72.7	78.5	83.9	86.6	89.0	87.0	86.2	82.6	70.6	62.8
	2%	WB	54.6	54.1	59.4	63.6	71.2	74.6	77.1	76.5	73.6	68.6	59.3	54.4
		MCDB	57.8	57.9	66.2	72.9	80.9	83.8	86.8	84.6	82.4	75.4	63.9	57.4
	5%	WB	49.4	48.4	56.5	61.4	69.2	73.2	75.6	74.9	71.7	65.1	56.1	50.4
		MCDB	52.3	53.1	62.0	70.1	78.4	82.1	84.0	82.0	79.5	71.4	61.3	53.7
	10%	WB	42.3	42.5	52.8	59.1	67.5	71.6	74.1	73.2	69.8	62.1	52.7	45.7
		MCDB	45.6	47.7	58.4	66.9	75.8	80.0	81.4	79.5	76.4	68.6	58.0	49.0
Mean Daily Temperature Range	5% DB	MDBR	15.6	17.2	19.3	22.5	21.6	20.9	20.0	21.3	23.2	22.2	19.3	14.5
		MCDBR	19.7	24.7	27.0	29.1	26.1	24.6	22.7	23.9	27.2	28.7	26.0	20.7
		MCWBR	17.1	19.8	17.0	16.2	12.7	10.8	10.3	11.1	12.6	15.5	17.9	17.6
	5% WB	MCDBR	18.7	22.9	23.3	23.3	22.1	20.4	19.7	19.2	22.0	23.7	21.1	19.4
		MCWBR	17.2	19.5	15.9	14.6	11.3	10.1	9.6	10.0	11.2	14.0	16.6	17.9
Clear-Sky Solar Irradiance	taub		0.309	0.329	0.355	0.390	0.430	0.451	0.462	0.450	0.394	0.364	0.326	0.305
	taud		2.437	2.359	2.359	2.282	2.200	2.181	2.165	2.183	2.319	2.394	2.470	2.472
	Ebn at Noon		272	281	285	280	270	264	259	259	268	264	263	264
	Edh at Noon		25	32	36	41	45	46	47	45	36	30	25	23
All-Sky Solar Radiation	RadAvg		509	770	1096	1485	1732	1893	1845	1712	1395	957	646	434
	RadStd		49	104	102	136	145	134	106	92	127	86	89	51

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (42 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+223	+122

Nomenclature: See separate page